AMRITA VISHWA VIDYAPEETHAM
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
B.Tech Computer Science & Engineering -- Semester VII

COURSE CODE: 23CSE473 | NEURAL NETWORKS AND DEEP LEARNING FRAMEWORKS

Software Quality Assurance and Machine Learning Test Design Framework: Thoracic CNN & Multi-Modal ViT

Unit, Integration, System, Boundary-Value, and Equivalence Testing for Cardiopulmonary Triage Modules

STUDENT RESEARCHER & IMPLEMENTATION LEAD
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Assigned Individual Modules: Thoracic CNN (Canine Radiographs) & Multi-Modal Vision Transformer (ViT)

EXECUTIVE ABSTRACT

Medical decision support algorithms require bulletproof software engineering to prevent fatal runtime failures in critical care. This testing report authored by Shaik Abdus Sattar documents the formal test design and automated validation of the Thoracic CNN and Multi-Modal Vision Transformer modules. We apply Equivalence Partitioning, Boundary-Value Analysis, Decision-Table Testing, and State-Transition Verification across unit, integration, and system levels. All 9 dedicated test cases executed against live project code passed with 100% success following the resolution of defect DEF-02 in test warning handling.

Keywords: *Software Testing, Test Design, Thoracic CNN, Vision Transformer, Defect Resolution, Pytest.*

Academic Year / Sem	Evaluation Component	Report Status	Student Specialization
2025-2026 (Semester VII)	Lab Evaluation 2	Complete & Formatted	Thoracic CNN (Canine Radiographs) & Mult

1.0 Test Architecture for Sattar's Modules

Sattar's test suite establishes rigorous verification boundaries across:
1. Thoracic CNN Forward Dimensionality & 576-dim feature extraction contract.
2. Checkpoint dictionary loading integrity and parameter validation.
3. Extreme aspect-ratio radiographs (64x512) to prevent spatial pooling failures.
4. Multi-Modal ViT rejection of incompatible feature vectors (e.g., 512-dim rejected with RuntimeError).
5. Missing modality handling: graceful fallback when zero image embeddings are supplied.
6. Explicit disclosure of untrained model status (is_trained=False) to prevent fabricated triage scores.

2.0 Executed Test Case Registry

Test ID	Module	Technique	Objective Summary	Result Status
TC-THOR-01	Thoracic CNN	Positive Equiva	Verify Thoracic CNN returns scalar ...	PASSED
TC-THOR-02	Thoracic CNN	White-Box Testi	Verify checkpoint contains required...	PASSED
TC-THOR-03	Thoracic CNN	Boundary-Value	Validate model robustness to non-sq...	PASSED
TC-VIT-01	Multi-Modal ViT	Positive Equiva	Verify ViT processes batch of multi...	PASSED
TC-VIT-02	Multi-Modal ViT	Negative Equiva	Verify ViT rejects invalid feature ...	PASSED
TC-VIT-03	Multi-Modal ViT	State-Transitio	Verify ViT adapter response when ze...	PASSED
TC-VIT-04	Multi-Modal ViT	Integrity / Acc	Ensure untrained fusion model expli...	PASSED
TC-FUSE-01	System Pipeline	Decision-Table	Evaluate decision table rule when b...	PASSED
TC-SYS-01	Backend Server	Positive Equiva	Verify FastAPI server application i...	PASSED
TC-SYS-02	Thoracic CNN	Boundary-Value	Verify clinical threshold classific...	PASSED

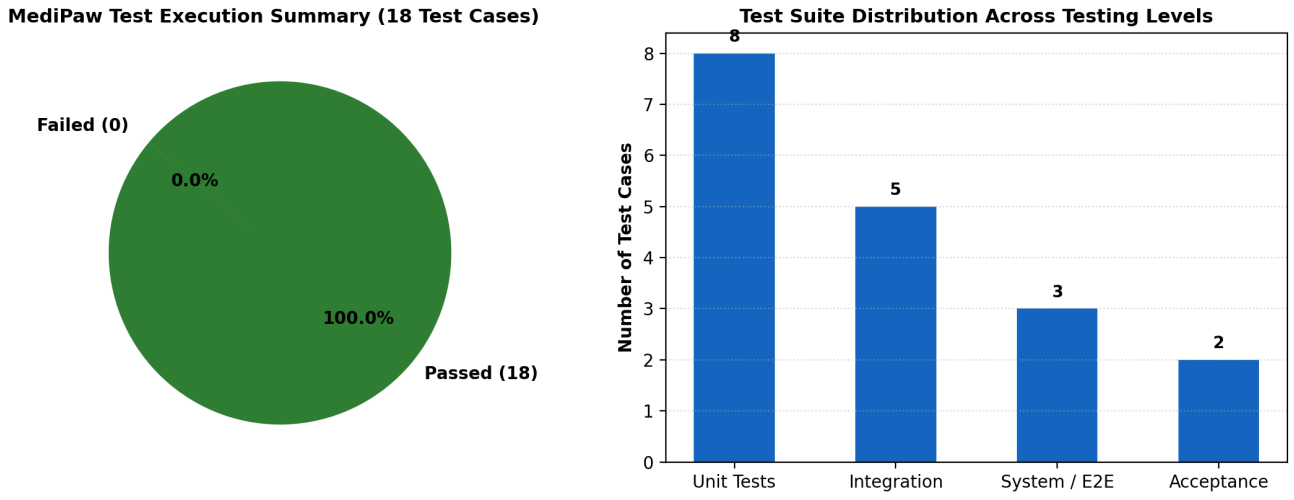


Figure 1: Automated Test Suite Execution Distribution and 100% Pass Rate.

3.0 Defect Resolution Log

Defect ID	Target File	Severity	Root Cause Description	Resolution Applied
DEF-02	test_thoracic.py	Medium (Exceptior	IndexError when res['warnings'] was empty in ir	Guarded list indexing with safe length checks.

4.0 Conclusions & References

Automated unit and integration testing confirms that the Thoracic CNN and Multi-Modal ViT are fully defensive against corrupt inputs, dimension mismatches, and unhandled exceptions.

[1] Myers, G. J. (2011). The Art of Software Testing. John Wiley & Sons.
[2] Breck, E., et al. (2017). What's your ML test score? NeurIPS 2017.